

EXTERNAL NOSE & BONY
BOUNDARIES OF NASAL
CAVITY

Learning Objectives.

At the end of lecture, students should be able to:

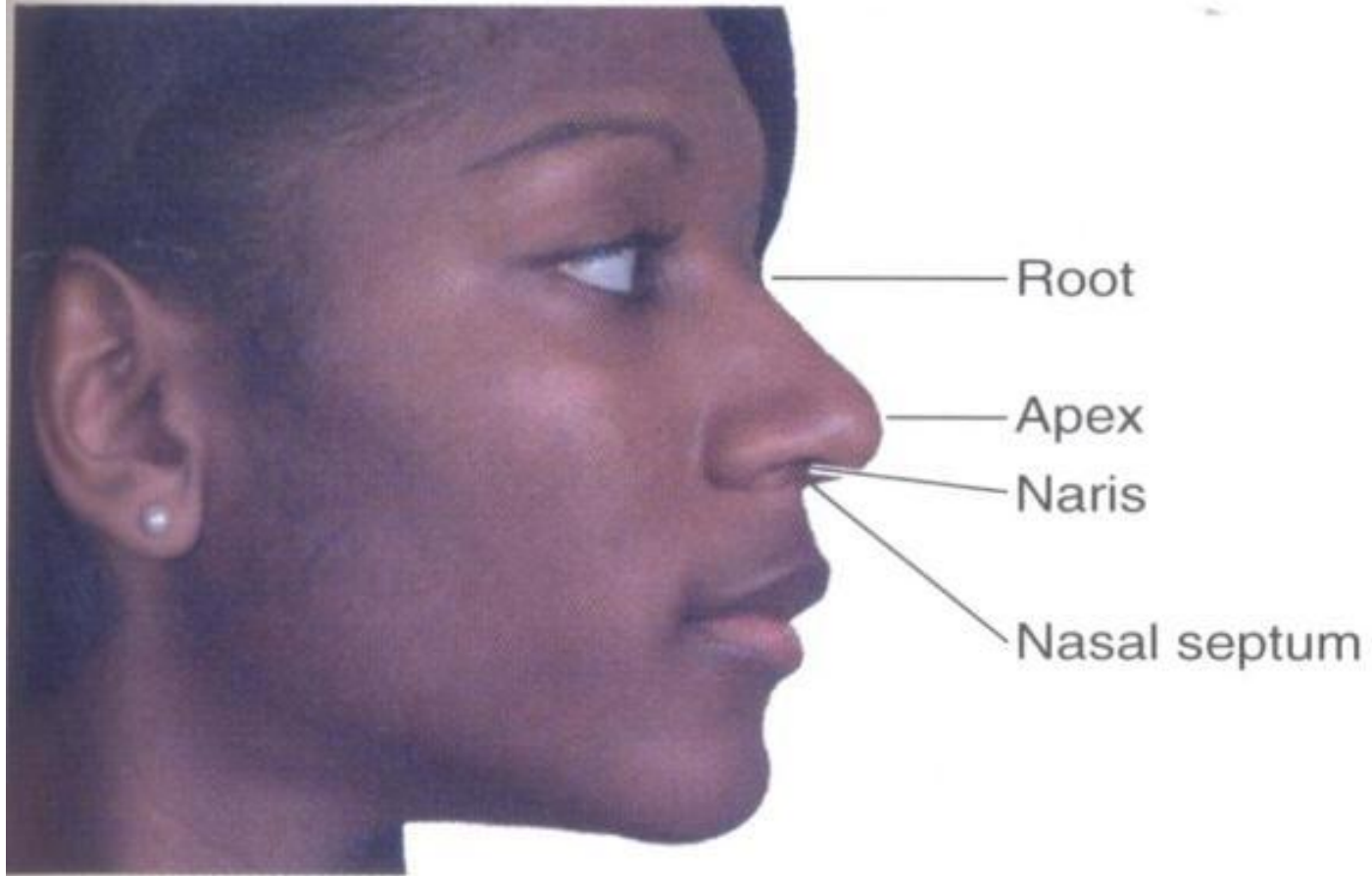
- Know the anatomy of external nose.
- Define the Boundaries of nasal cavity.
- Know Blood vessels of nose.
- Know Nerve supply of nose.

Nose:

- Organ of respiratory passage.
- Organ of smell
- Receptors for smell lies in upper 1/3rd of nose covered by olfactory mucosa and or factory cells.
- Rest of nasal cavity lined by respiratory mucosa.

Functions:

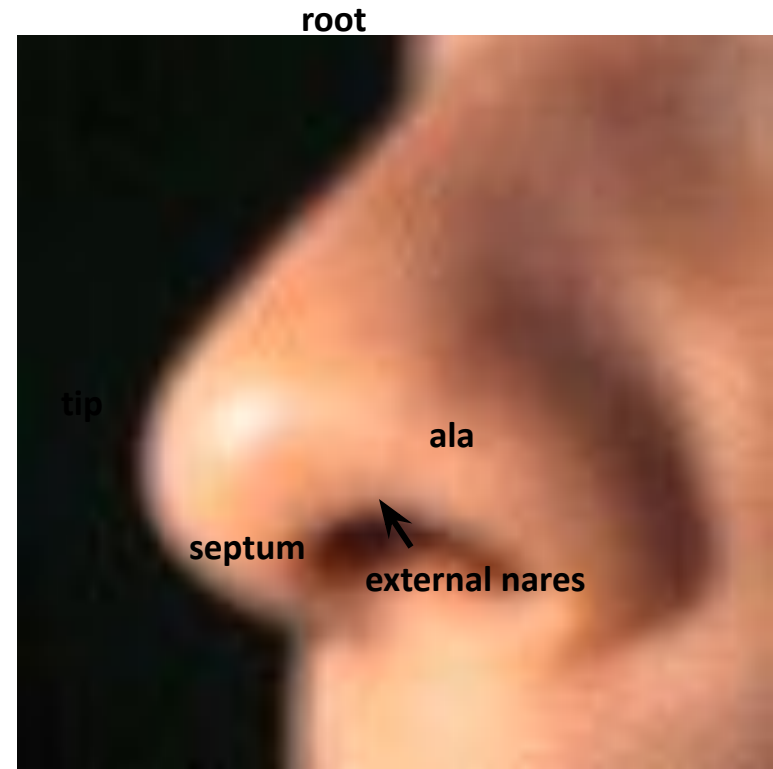
- serous glands to make the air moist
- mucous glands to trap the dust and foreign particles.



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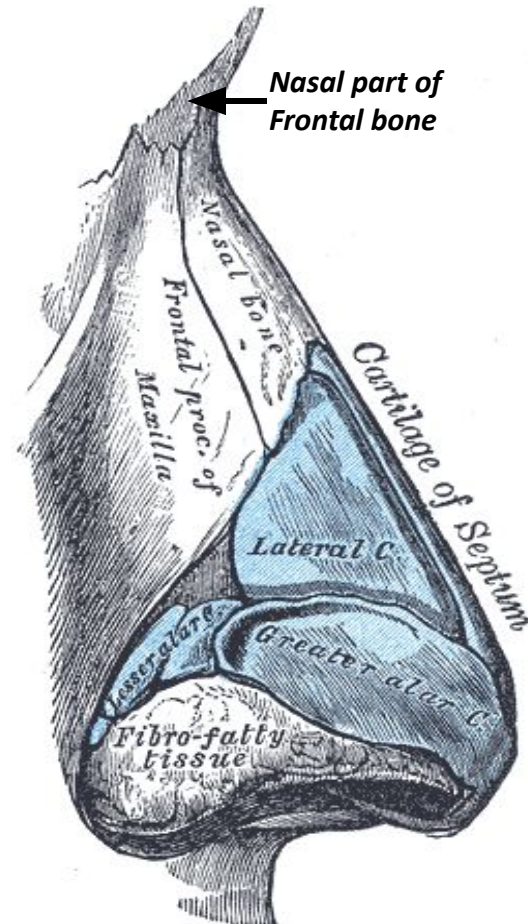
Nose

- Only externally visible part of the respiratory system
- Has a free tip and is attached to the forehead by the root or the bridge
- Has two openings, the anterior (external) nares or nostrils, which lead to the nasal cavity
- Each nostril is bounded laterally by the ala and medially by the nasal septum



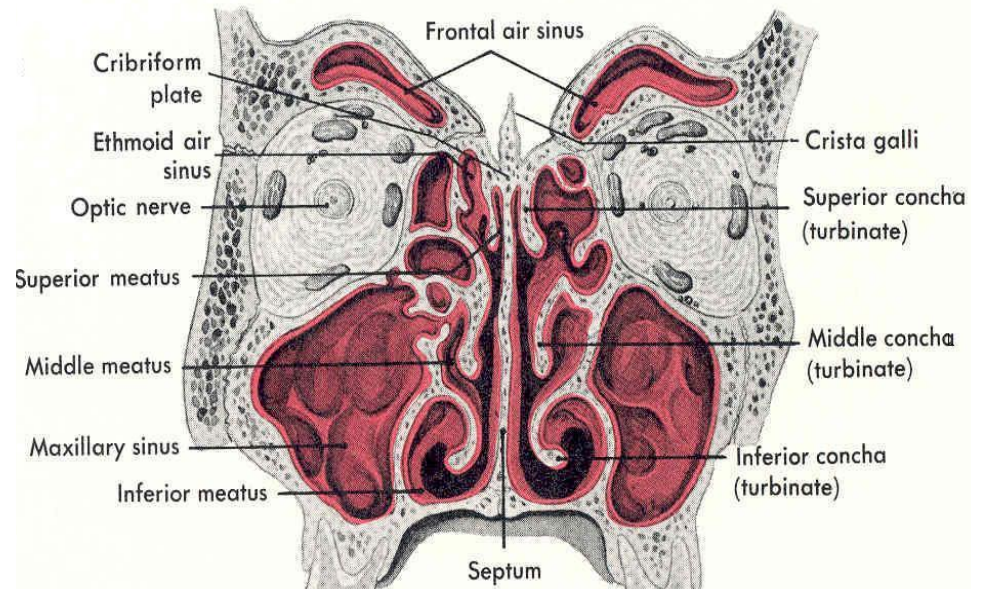
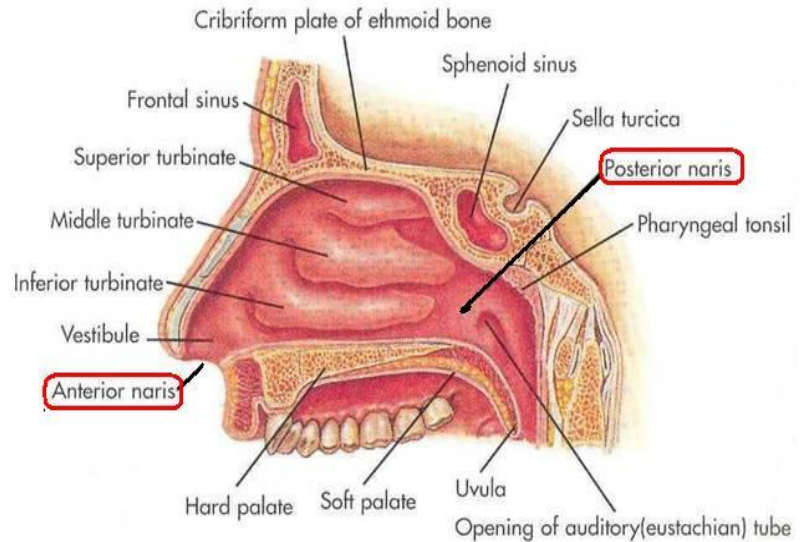
Nose: Structure

- Nose consists of bony & cartilaginous framework
- Formed above by the:
 - Nasal bones
 - Frontal processes of maxilla
 - Nasal part of frontal bone
- Formed below by plates of hyaline cartilage, which include upper & lower nasal cartilages and the septal cartilage

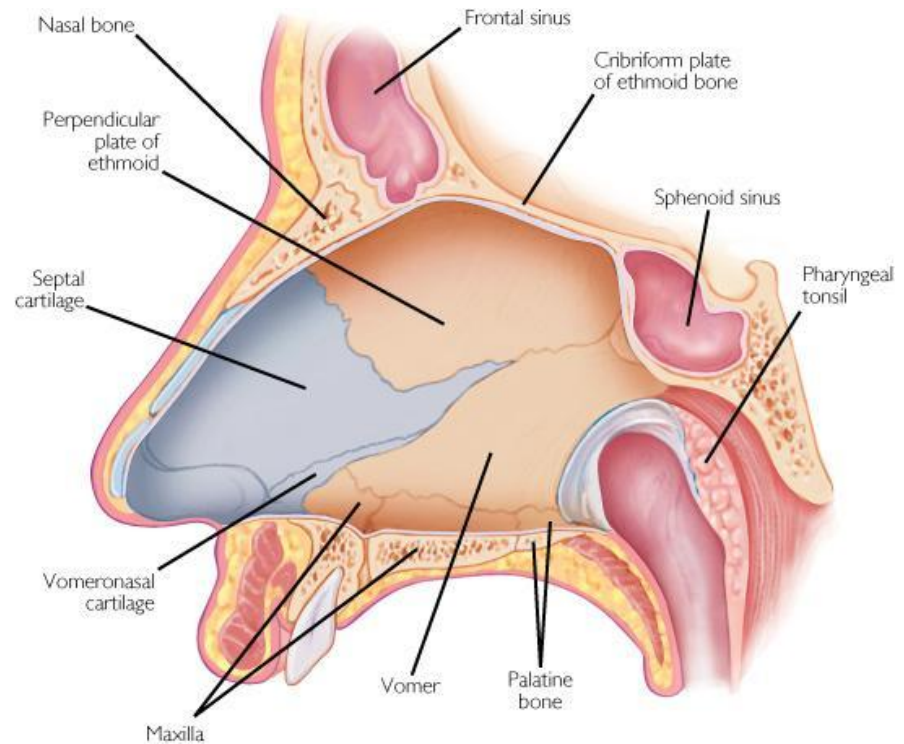


Nasal Cavity

- Extends from the external (anterior) nares to the posterior nares (choanae)
- Divided into right & left halves by the nasal septum
- Each half has a:
 - Floor
 - Roof
 - Lateral wall
 - Medial wall (septum)

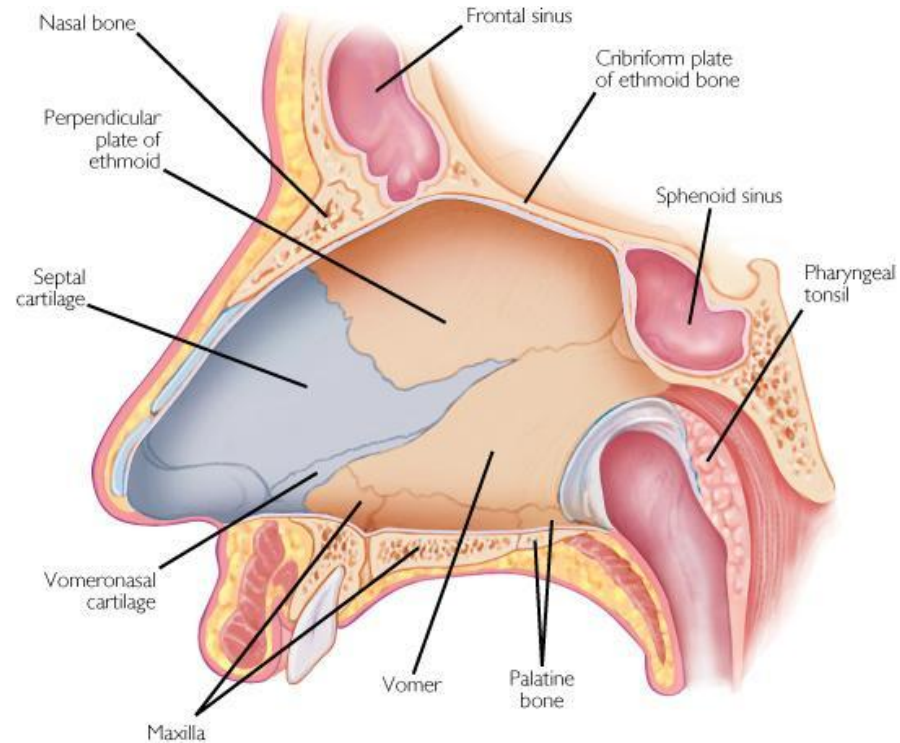


- **Roof**
- Is narrow & formed (from behind forward) by the:
 - Body of sphenoid
 - Cribriform plate of ethmoid bone
 - Frontal bone
 - Nasal bone & cartilage
- **Floor**
 - Separates it from the oral cavity
 - Formed by the hard (bony) palate



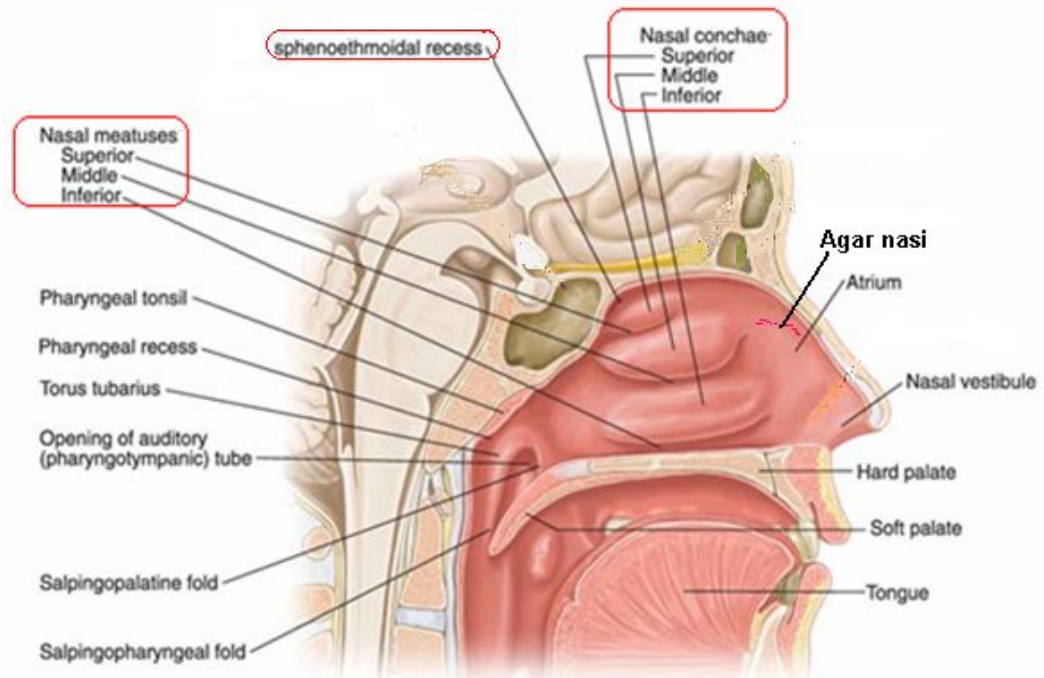
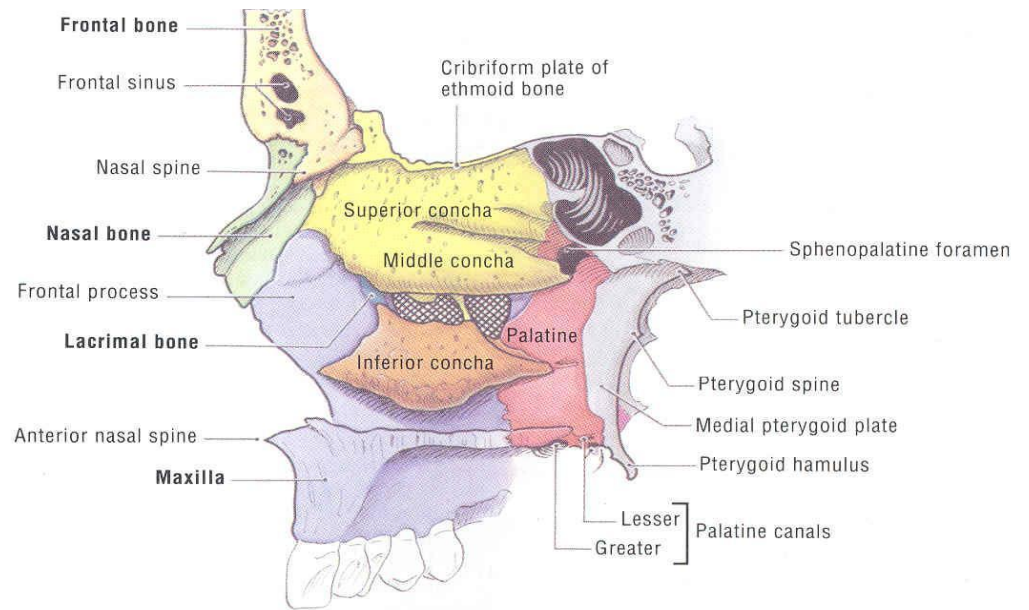
Medial Wall (Nasal Septum)

- Osteocartilaginous partition, only rarely lying in the midline
- Covered by the mucoperiosteum
- Formed:
 - Superiorly by the vertical (perpendicular) plate of ethmoid bone
 - Posteriorly by the vomer bone
 - Anteriorly by the septal cartilage

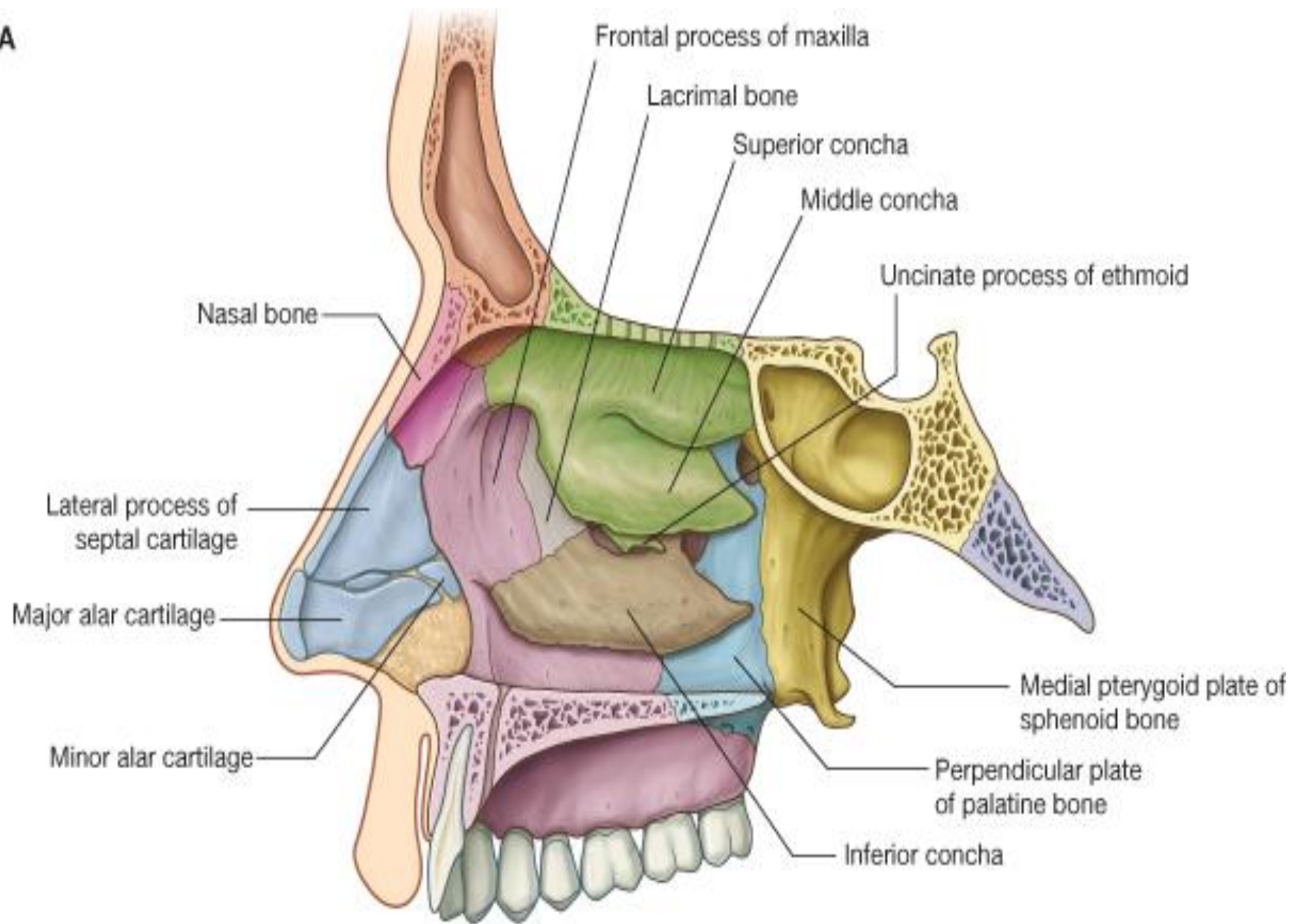


Lateral Wall

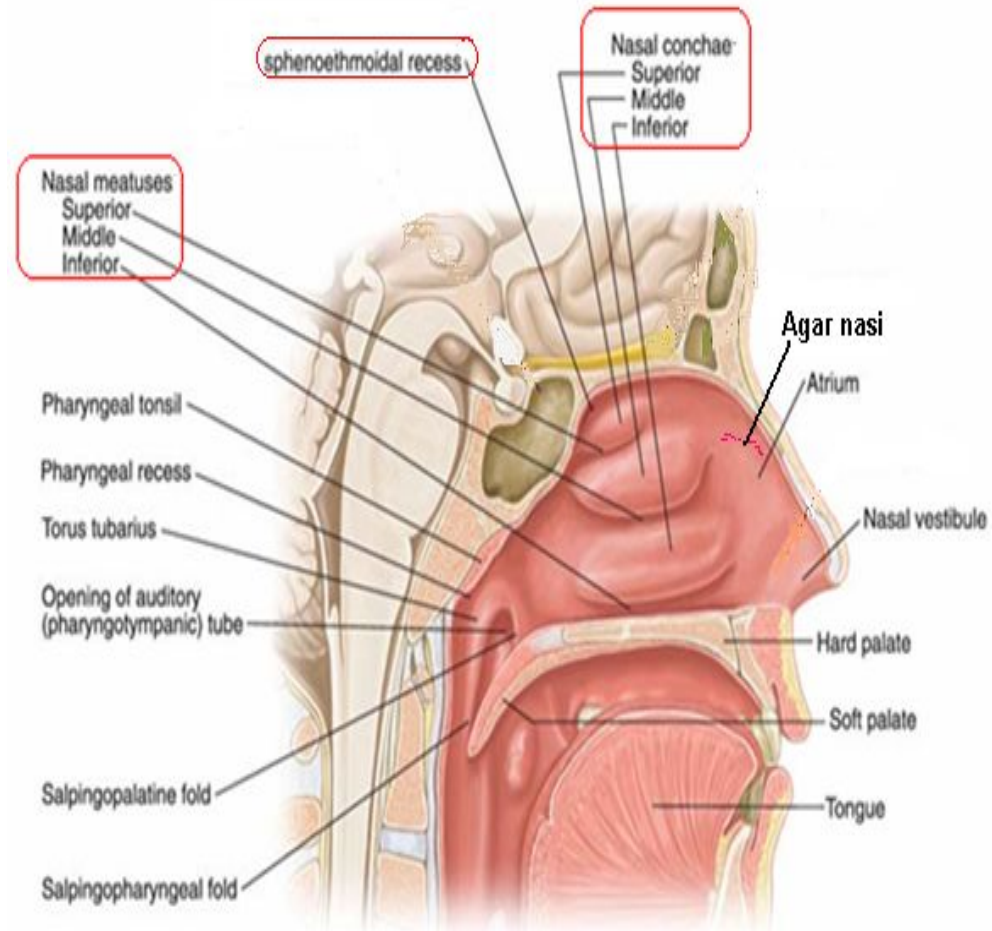
Shows three horizontal bony projections, covered by mucous membrane, the superior, middle & inferior conchae (turbinates)



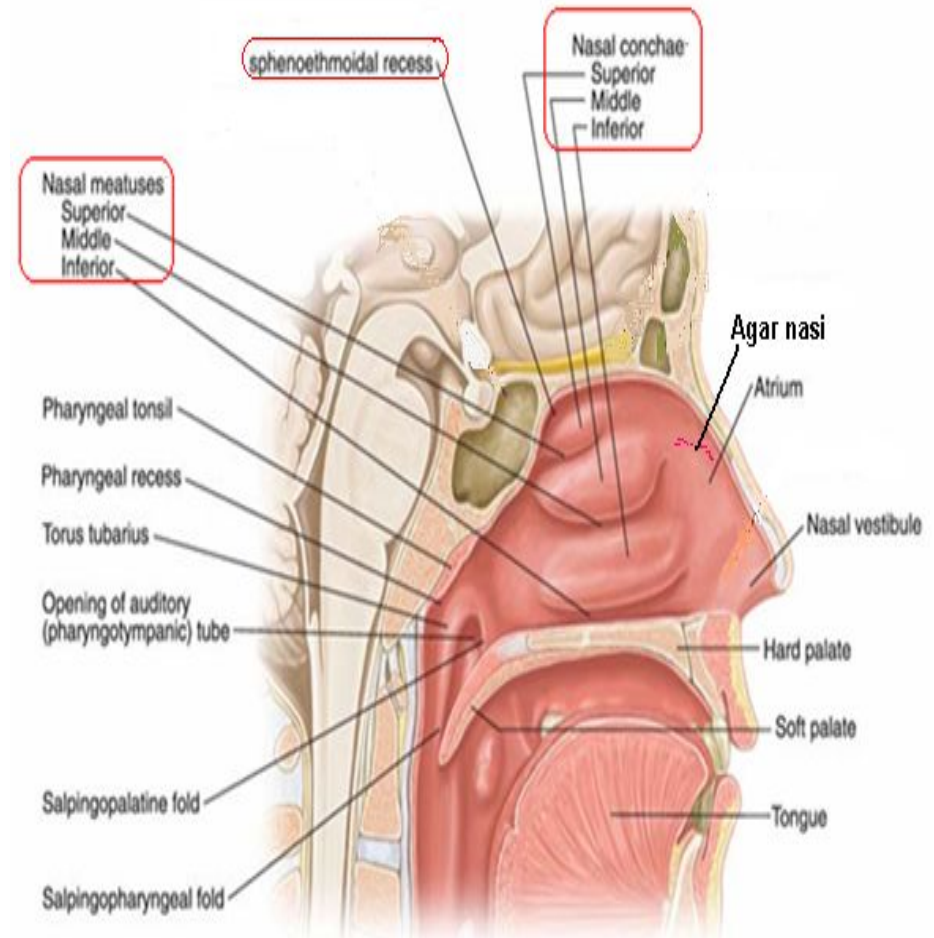
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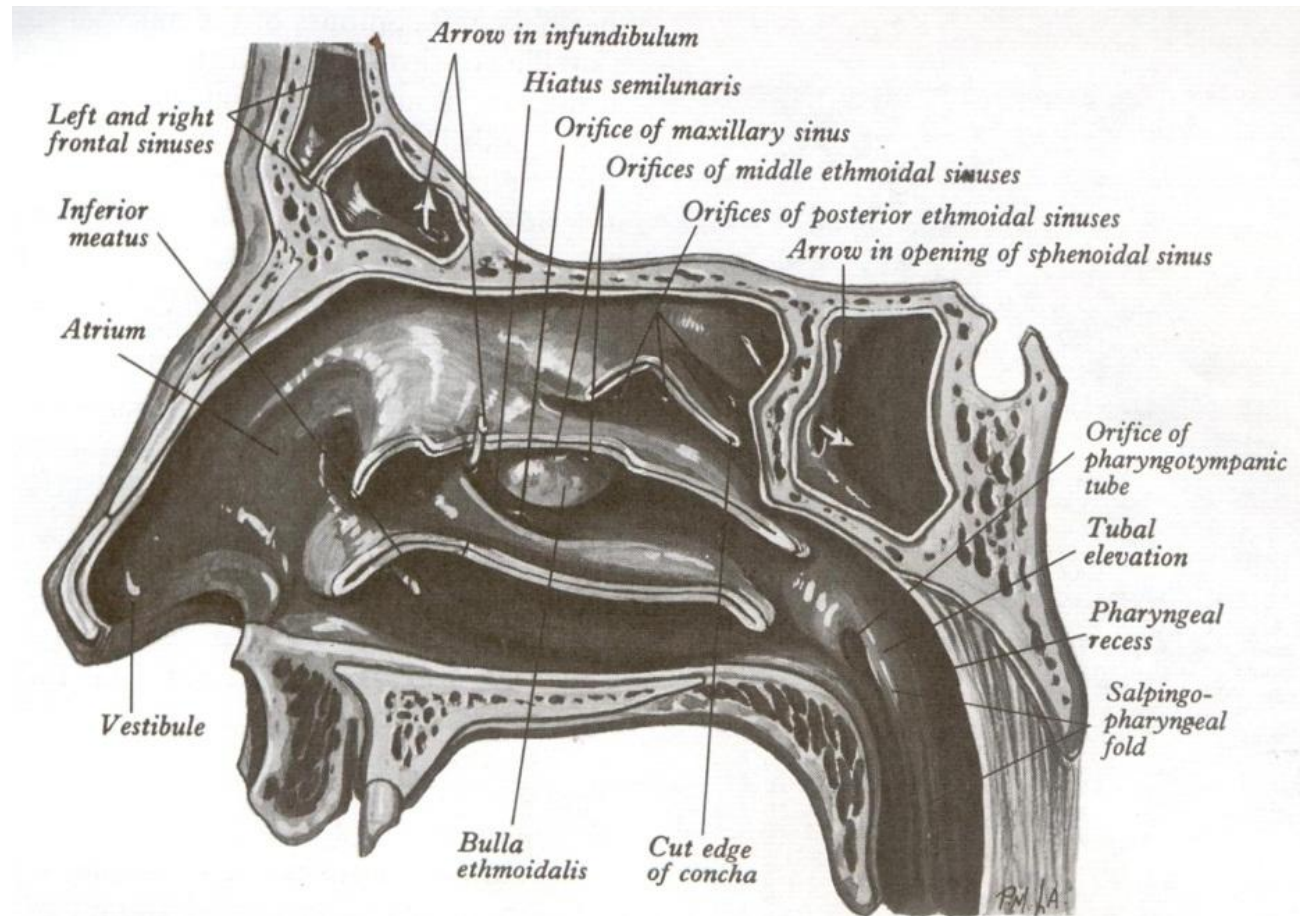
- The superior and middle conchae are parts of the ethmoid bone, whereas the inferior concha is a separate bone.
- The cavity below each concha is called a meatus and are named as superior, middle & inferior corresponding to the conchae



- The small space above the superior concha is called the sphenothmoidal recess
- The middle meatus is continuous in front with a depression called the atrium



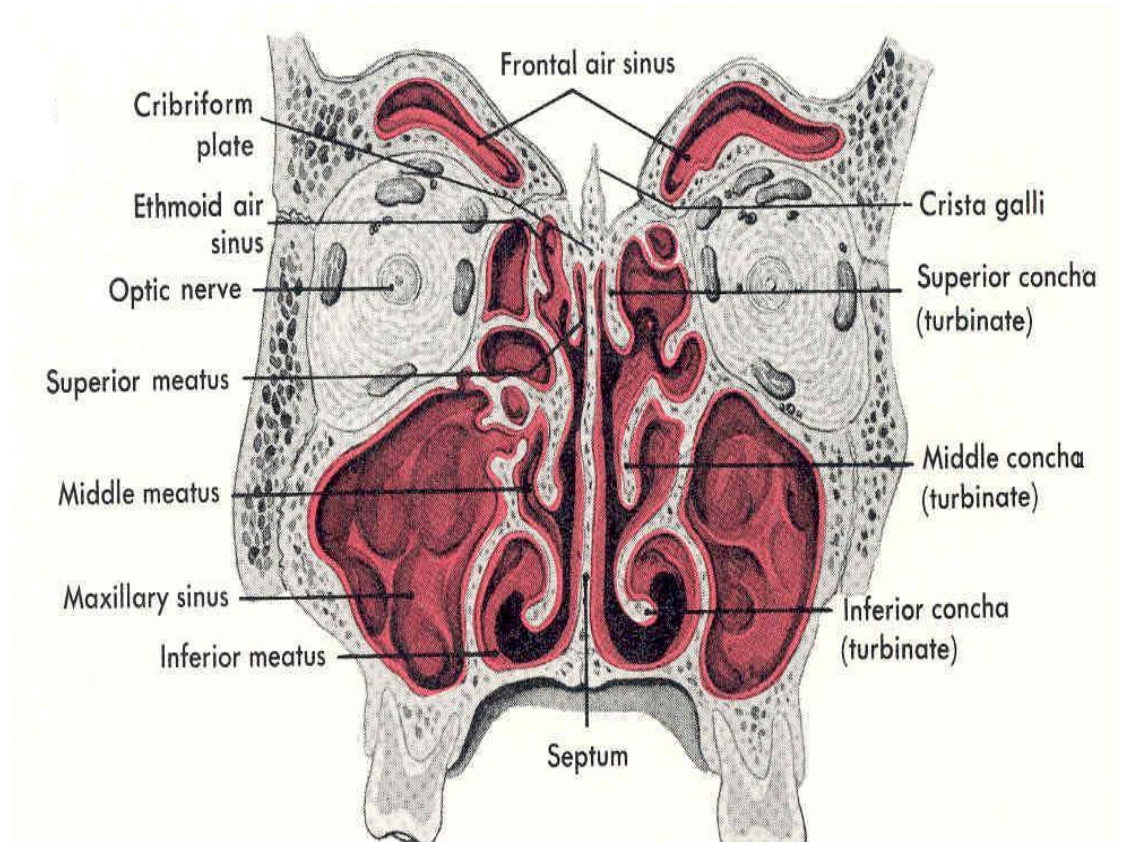
- Atrium is limited above by a ridge called agar nasi
- Below and in front of atrium, and just within the nostril lies the vestibule



The conchae increase the surface area of the nasal cavity

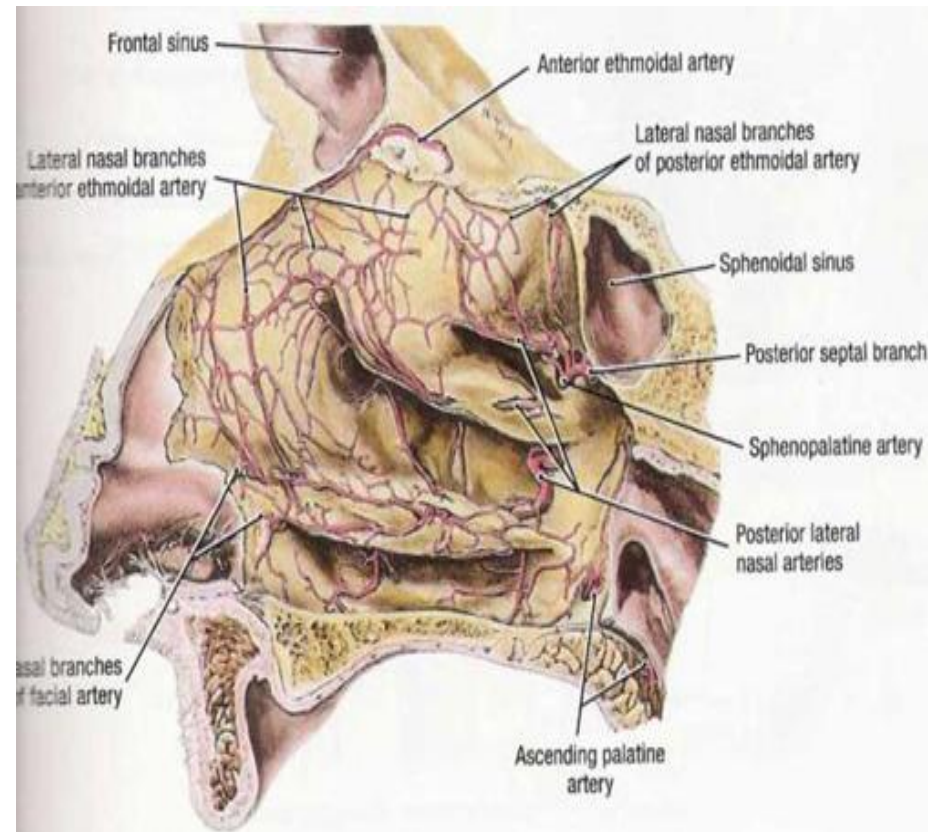
The recess & meati receive the openings of the:

- Paranasal sinuses
- Nasolacrimal duct



Arterial supply of nose

- The arterial supply of medial and lateral walls of nasal cavity is from five sources:
 1. Anterior ethmoidal artery (from ophthalmic artery).
 2. Posterior ethmoidal artery (from ophthalmic artery).
 3. Sphenopalatine artery (from maxillary artery).



4. Greater palatine artery
(from maxillary artery).

5. superior labial artery
(from facial artery).

Venous supply of nose

Submucosal venous
plexus.



Nerve supply of nose

- Sensory – (Trigeminal nerve).
- **Anterior superior** – Anterior-ethmoidal nerve (ophthalmic nerve).
- **Anterior inferior** – Anterior superior alveolar nerve (maxillary nerve).
- **Posterior superior** – Posterior superior, lateral nasal (maxillary nerve).

Self Assessment

- Cartilaginous framework of nose is formed by?
- Name the Boundaries of Medial wall (nasal septum)?
- Name the arteries supplying the nose?